

enforcing the standard: tool-based code review

SCRUB

The screenshot displays the SCRUB static analysis tool interface. The top window shows the source code for `mrf_mrb_list.c` with line numbers 0294 to 0317. The code includes a function `return 0;` and a function `MrB MrBList_Next` that uses a mutex. The bottom window shows the analysis results, with a list of issues and a summary table.

static analysis tools
run as additional mechanical
reviewer demons in the
background (all the time)

files	comments	save	analyses:	stats	gcc	gcc_strict	coventy	codesonar	uno	msl
scandore0070 <High> <code>mrf_mrb_list.c:305:</code> scandore0070 <High> Does this require a lock? Verify if the function is only called from within MRF context behind an I FACE. If so, then the lock can be removed. david1001 <Disagree> Yes. MrBList is used for interprocess communication, as lpcQ normally is used. MrBList is used for IPC between the actors that act upon DPs. MrBList is used because the actors must act upon a set of DPs over a possibly extended period of time, rather than one-at-a-time as with lpcQ. cichy2069 <No Action Required> cichy0063 <High> <code>mrf_mrb_list.c:487:</code> cichy0063 <High> What is this? We need to discuss this - I'm not okay with doing extra checking in WSTS david1002 <Disagree> This function (MrBList_Check) can only be safely executed during UTH. At the point of invocation, within <code>mrf_to_mrb_map.c</code>, <code>MrFile_CheckOppressive()</code>, a check is made to ensure that the MRF is running single threaded in UTH. Therefore, this is not a check that is only performed in normal multi-threaded WSTS execution.										

Sort by
◆ Name
◆ Importance
◆ ToAnswer
Agrees: 117
Disagrees: 15
Discuss: 39
ToAnswer: 4
ToResolve: 4

G.J. Holzmann , “**Scrub: a tool for code reviews**,”
Innovations in Systems and Software Engineering, 2010, Vol. 6, Nr. 4, pp. 311-318.